Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

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Grade and Subject Teacher's Signature	

Task Number	10
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Continuous Testing in Jenkins: Integrate the Loan Application Tracking Portal Selenium suite into Jenkins. Publish test reports and configure the pipeline so failed tests stop deployment. Correct at least one deliberately introduced defect and rerun the pipeline.
Deliverable:	

Step-by-Step Quality Gate & Defect Injection Guide

1: Push the Quality-Gated `Jenkinsfile` (Updated `Jenkinsfile`)

```
D:\Vicky_Project\DevOps\DevOps_Project>git add Jenkinsfile
warning: in the working copy of 'Jenkinsfile', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "ci: add dedicated Selenium quality gate stage with test reporting
and screenshot archiving"
[main b2f6fa5] ci: add dedicated Selenium quality gate stage with test reporting and screenshot archiving
 1 file changed, 26 insertions(+), 14 deletions(-)

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 899 bytes | 899.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   9b103a5..b2f6fa5  main -> main
```

2: Introduce a Deliberate Defect (To Fail the Quality Gate)

Replace line form:

<h3 class="mb-3 text-center text-primary" id="pageTitle">Submit Loan Application</h3>

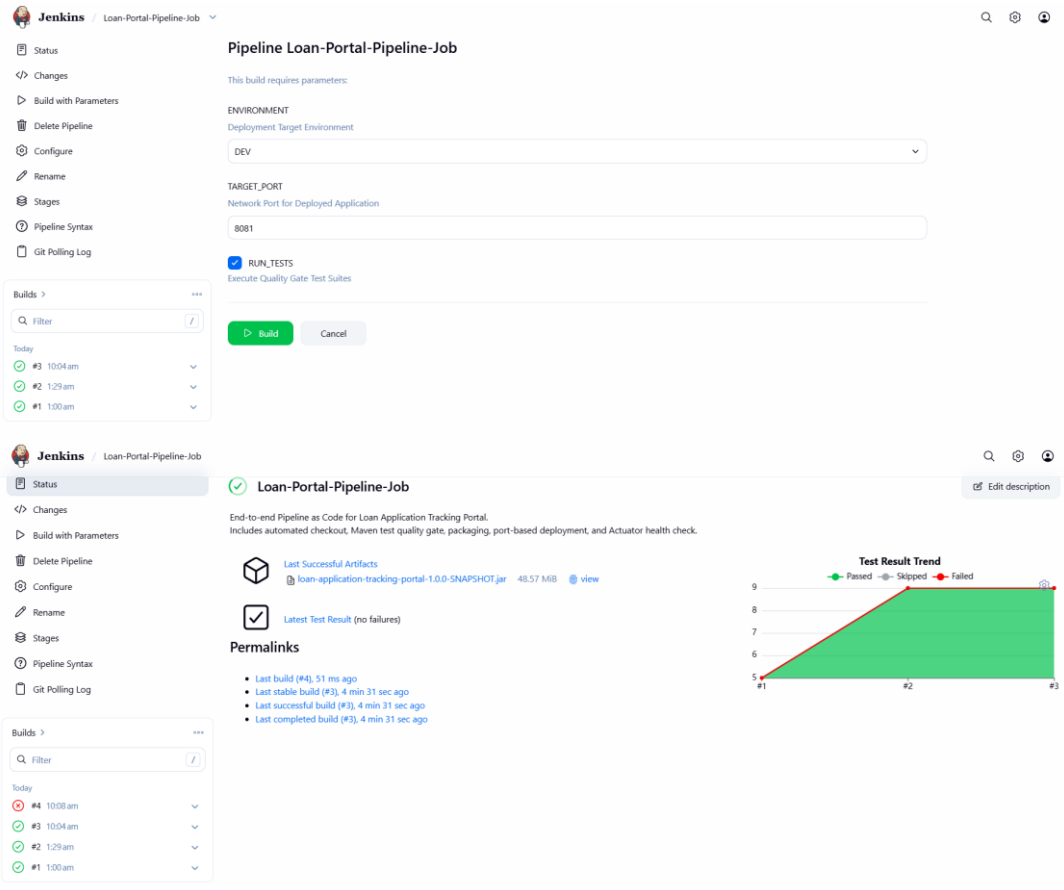
To:

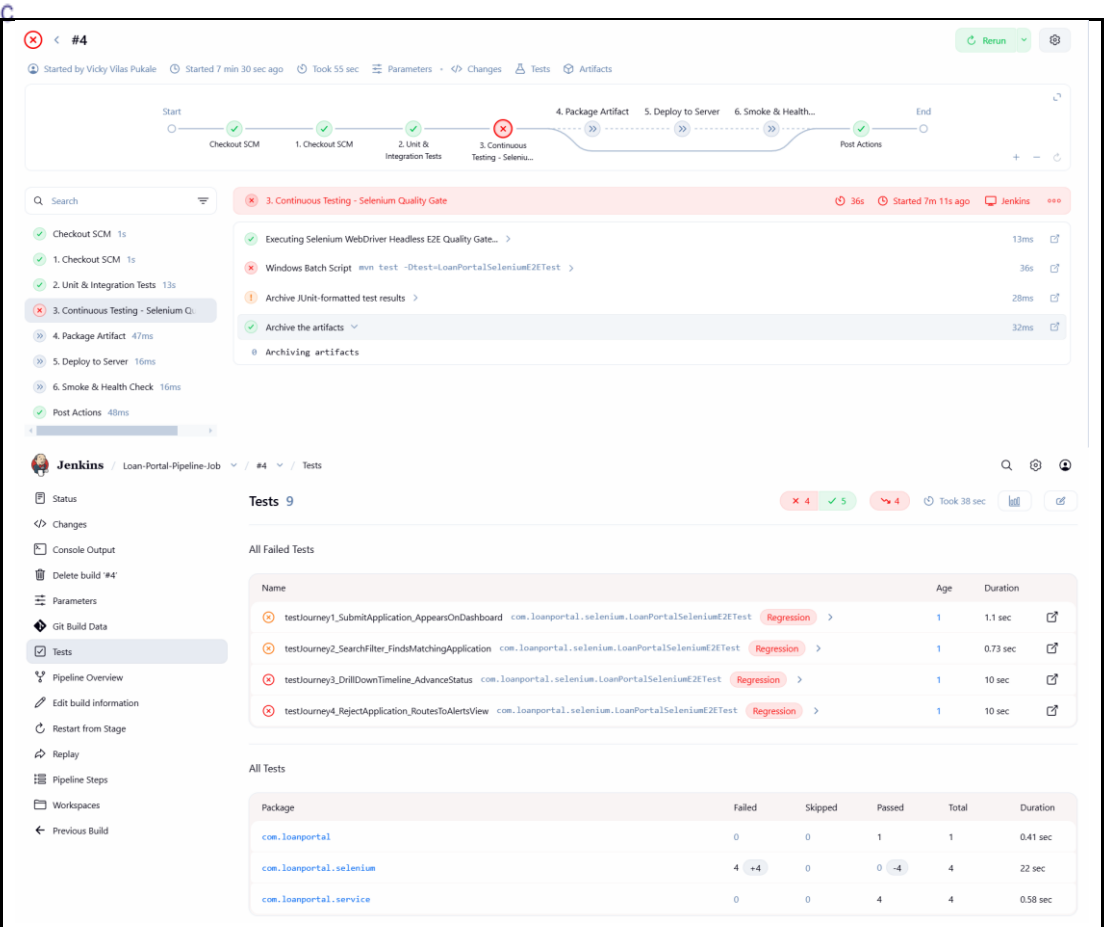
<h3 class="mb-3 text-center text-primary" id="pageTitle">BROKEN Loan Application Page</h3>

```
D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "feat(ui): alter page header (deliberate defect injection for QA ga
te testing)"
[main 581d7c6] feat(ui): alter page header (deliberate defect injection for QA gate testing)
 1 file changed, 1 insertion(+), 1 deletion(-)

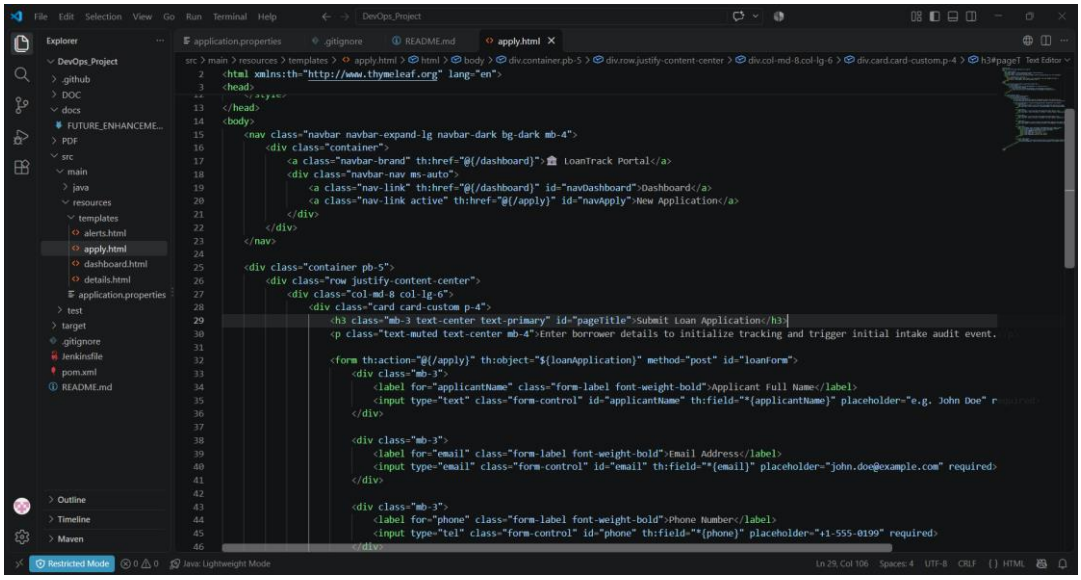
D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 686 bytes | 686.00 KiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   b2f6fa5..581d7c6  main -> main
```

3: Run the Pipeline & Observe the Quality Gate Failure:





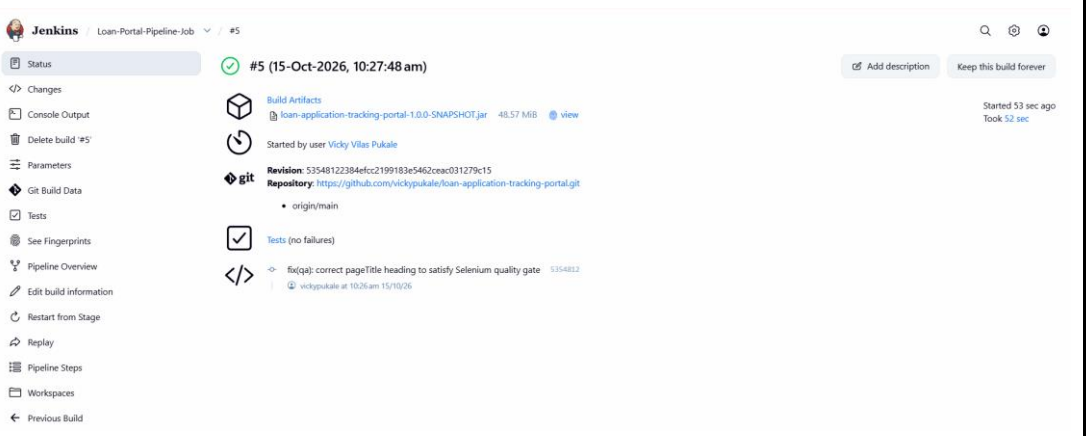
4: Correct the Defect & Verify Successful Recovery:

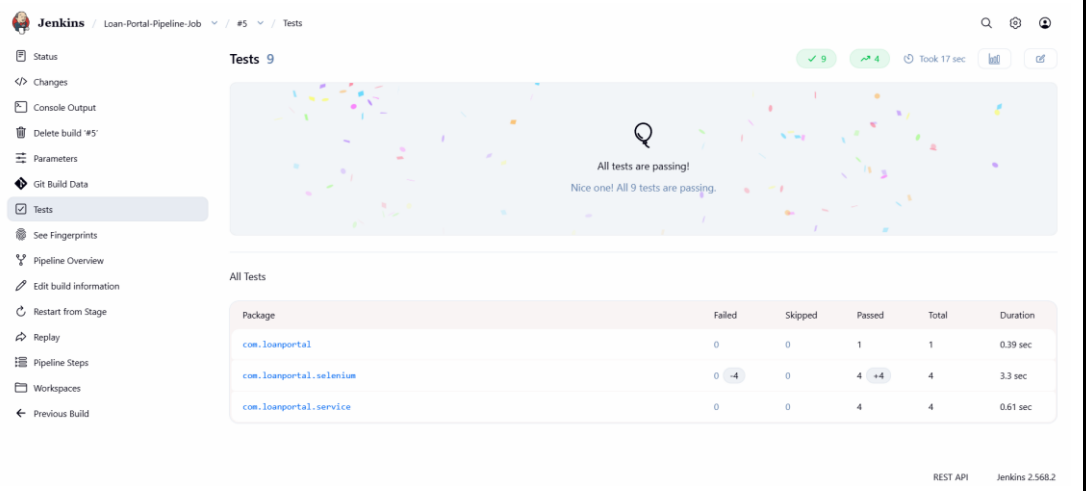


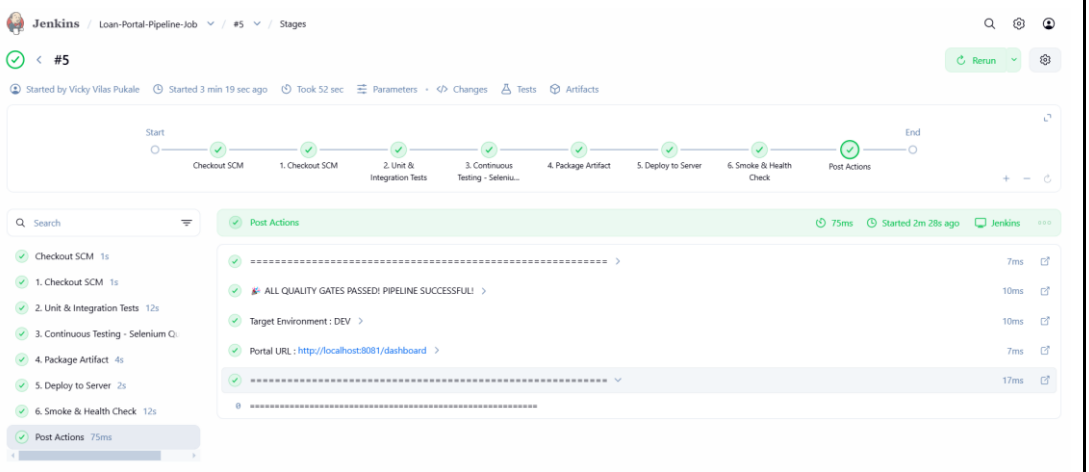

```
D:\Vicky_Project\DevOps\DevOps_Project>git add src/main/resources/templates/apply.html

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "fix(qa): correct pageTitle heading to satisfy Selenium quality gat
e"
[main 5354812] fix(qa): correct pageTitle heading to satisfy Selenium quality gate
 1 file changed, 1 insertion(+), 1 deletion(-)

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 677 bytes | 677.00 KiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   581d7c6..5354812  main -> main
```


Jenkins / Loan-Portal-Pipeline-Job / #5
Status
Changes
Console Output
Delete build '#5'
Parameters
Git Build Data
Tests
See Fingerprints
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#5 (15-Oct-2026, 10:27:48 am)
Add description
Keep this build forever
Build Artifacts
loan-application-tracking-portal-1.0.0-SNAPSHOT.jar 48.57 MiB view
Started 53 sec ago
Took 52 sec
Started by user Vicky Vilas Pukale
Revision: 53548122384efcc2199183e5462ceac031279c15
Repository: https://github.com/vickypukale/loan-application-tracking-portal.git
origin/main
Tests (no failures)
fix(qa): correct pageTitle heading to satisfy Selenium quality gate
vickypukale at 10:26 am 15/10/26


Jenkins / Loan-Portal-Pipeline-Job / #5 / Tests
Tests 9
Took 17 sec
All tests are passing!
Nice one! All 9 tests are passing.
All Tests
Package
Failed
Skipped
Passed
Total
Duration
com.loanportal
com.loanportal.selenium
com.loanportal.service
REST API
Jenkins 2.568.2


Jenkins / Loan-Portal-Pipeline-Job / #5 / Stages
#5
Rerun
Started by Vicky Vilas Pukale
Started 3 min 19 sec ago
Took 52 sec
Parameters
Changes
Tests
Artifacts
Start
Checkout SCM
1. Checkout SCM
2. Unit & Integration Tests
3. Continuous Testing - Seleniu...
4. Package Artifact
5. Deploy to Server
6. Smoke & Health Check
Post Actions
End
Search
Post Actions
75ms
Started 2m 28s ago
Jenkins
Checkout SCM 1s
1. Checkout SCM 1s
2. Unit & Integration Tests 12s
3. Continuous Testing - Selenium
4. Package Artifact 4s
5. Deploy to Server 2s
6. Smoke & Health Check 12s
Post Actions 75ms
ALL QUALITY GATES PASSED! PIPELINE SUCCESSFUL!
Target Environment : DEV
Portal URL : http://localhost:8081/dashboard